Open in app ↗

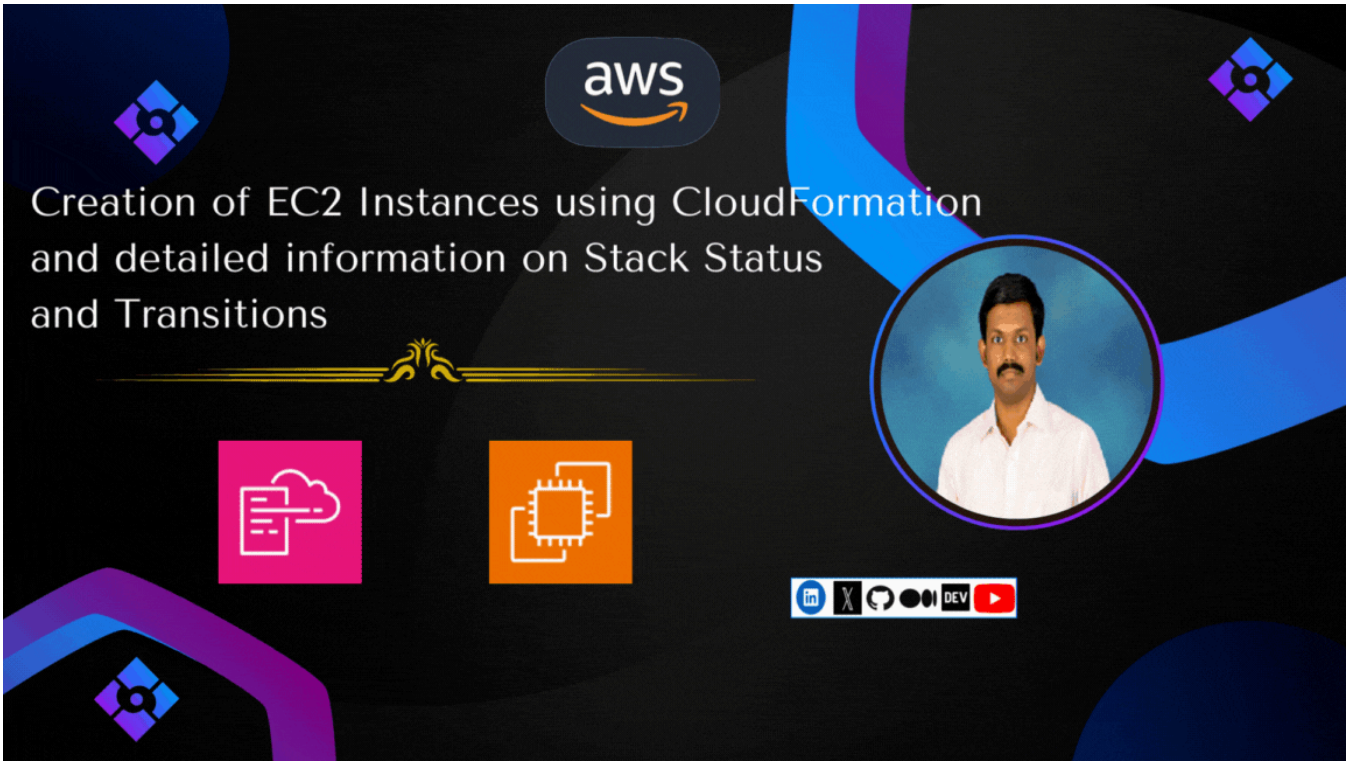
Medium Search

Creation of EC2 Instances using CloudFormation and detailed information on Stack Status and Transitions

5 min read · Jan 1, 2024

Vijayaraghavan Vashudevan Follow

Listen Share ••• More



Banner

🥷 Table of Contents:

- 📚 Synopsis:
- 📋 Prerequisites:
- 🌈 Creating an EC2 Instance using CloudFormation:
- 🧹 Clean Up Instructions to avoid billing:
- 🌈 Important to understand stack status and transitions:
  - 🌈 Create Stack Status Transitions:
  - 🌈 Update Stack Status Transitions:
  - 🌈 Delete Stack Status Transitions:
- 🌈 Summary:

📚 Synopsis:

📌 In this blog post, We will explore how to create the EC2 instances using the CloudFormation template as well as information about stack status and the transitions that occur as the stack moves from one status to another.

📋 Prerequisites:

📌 AWS account with admin privileges

📌 Internet connection

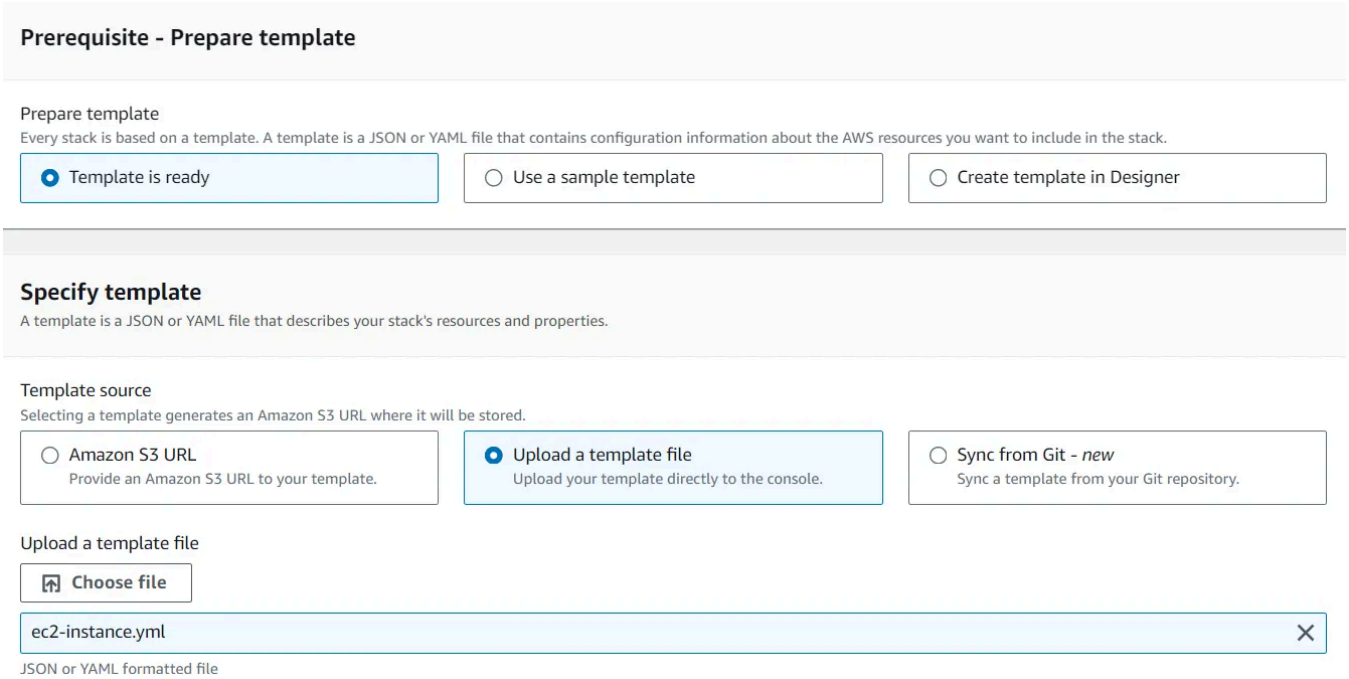
🌈 Creating an EC2 Instance using CloudFormation:

📌 We will be using the below template to create an EC2 Instance by replacing the following defaults of SubnetId, SecurityGroupId, and ImageId with actual values that exist

```
Parameters:
  SubnetId:
    Type: String
    Default: add-your-value-here
  SecurityGroupId:
    Type: String
    Default: add-your-value-here
  ImageId:
    Type: String
    Default: add-your-value-here
```

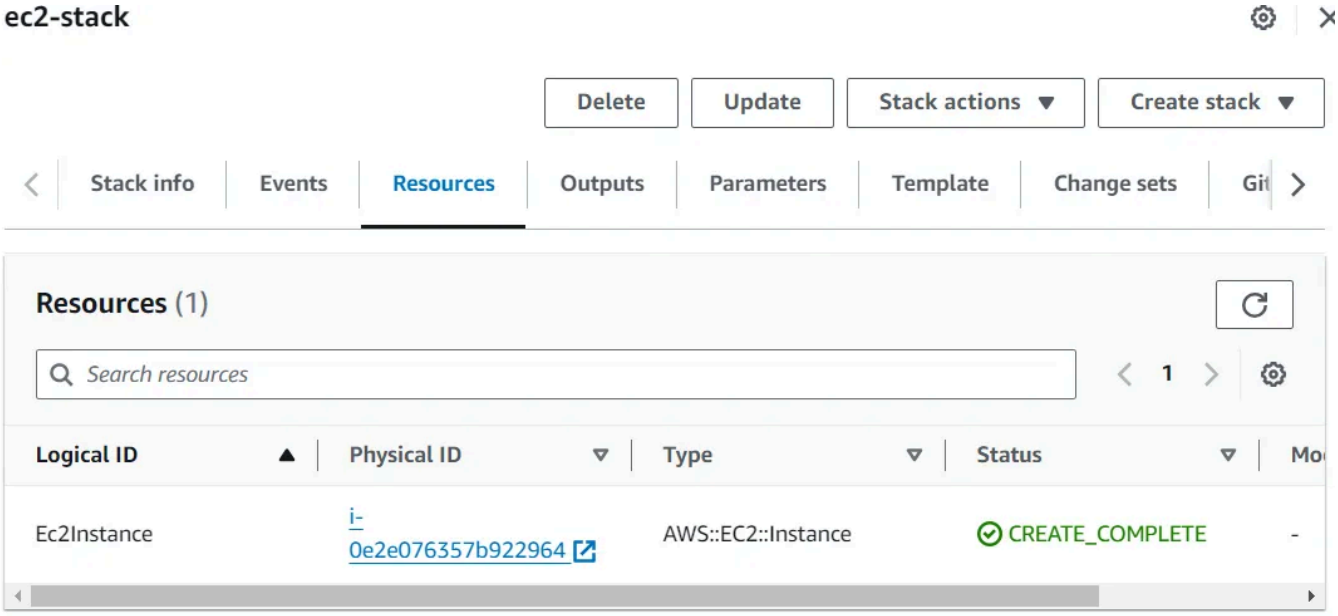
```yaml
Resources:
  Ec2Instance:
    Type: AWS::EC2::Instance
    Properties:
      InstanceType: t3.micro
      ImageId:!Ref ImageId
      SubnetId:!Ref SubnetId
      SecurityGroupIds:
      - !Ref SecurityGroupId
```

📌 Go to the management console, select the CloudFormation stack, and click on Create Stack. Click Next and give the stack name as ec2-stack. Click on Submit.

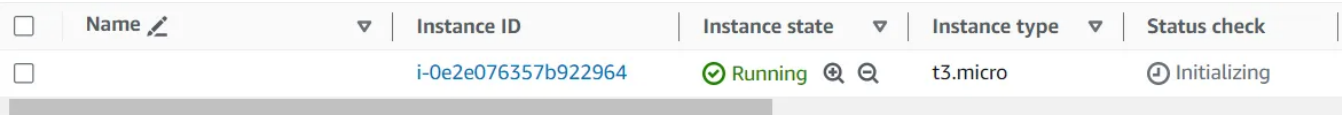


Stack creation

📌 Once the stack has been created, You can see the EC2 Instance under the resources section below



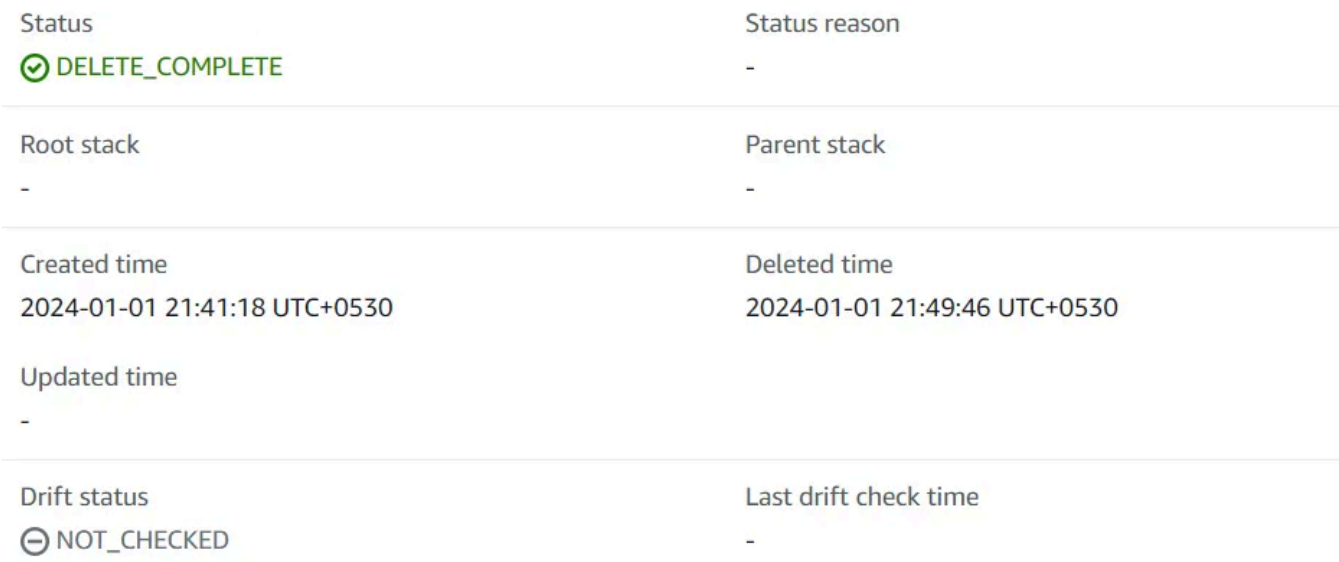
Resource



EC2 Instance

🧹 Clean Up Instructions to avoid billing:

♻️ To avoid hourly charges, **remember to delete this stack afterward**, by going to the CloudFormation stack section in the UI and selecting the **Delete** option.



Delete Stack

🌈 Important to understand stack status and transitions:

📌 Will discuss in detail status transitions for three scenarios: creation, update, and deletion of stacks

For each scenario, we will use the following color code:

➡️ **Gray for intermediate statuses.** You should just wait for the next transition.

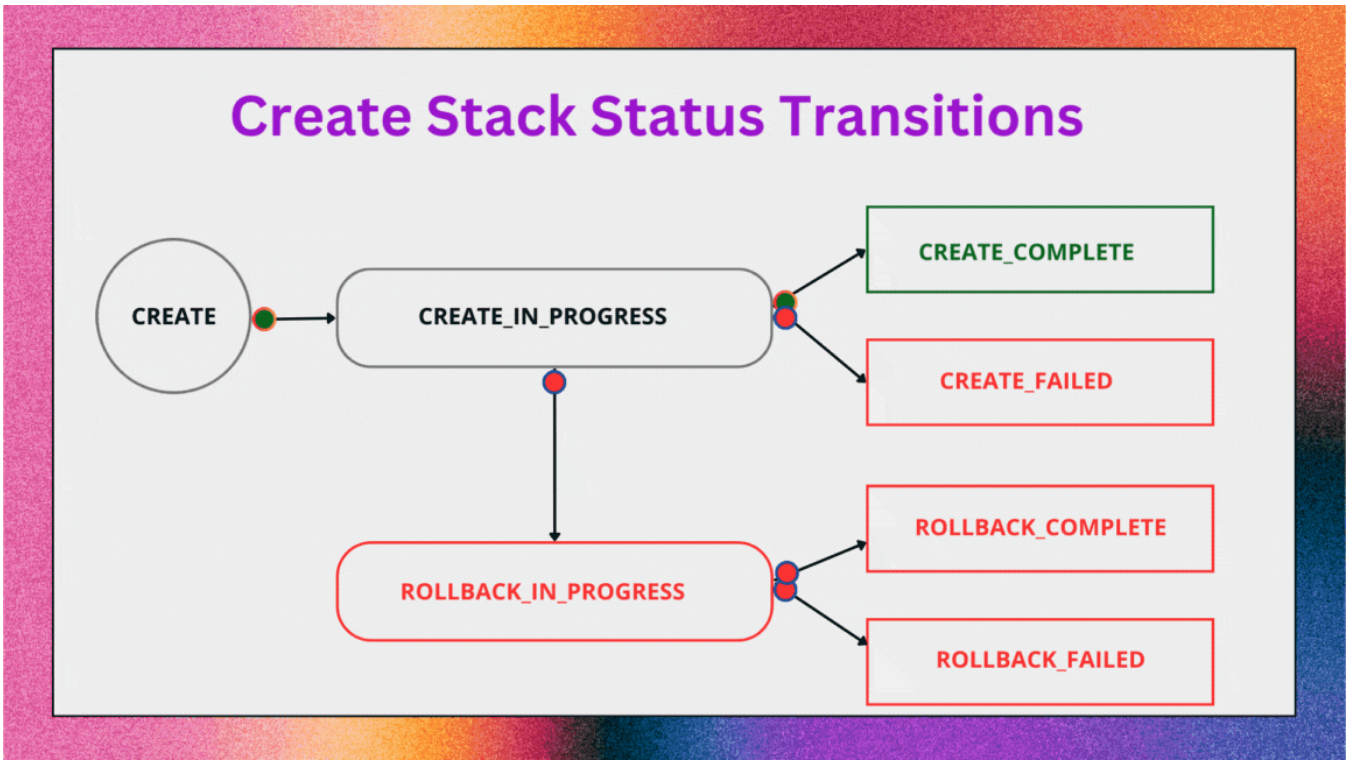
➡️ **Green for successful end statuses.** Your changes are applied, and everything should be good.

➡️ **Blue for successful rollbacked states**, which will require you to review the Events and correct issues before retrying.

➡️ **Red for failed states**, which will require you to manually correct issues or, in the worst-case scenario, delete the stack.

🌈 Create Stack Status Transitions:

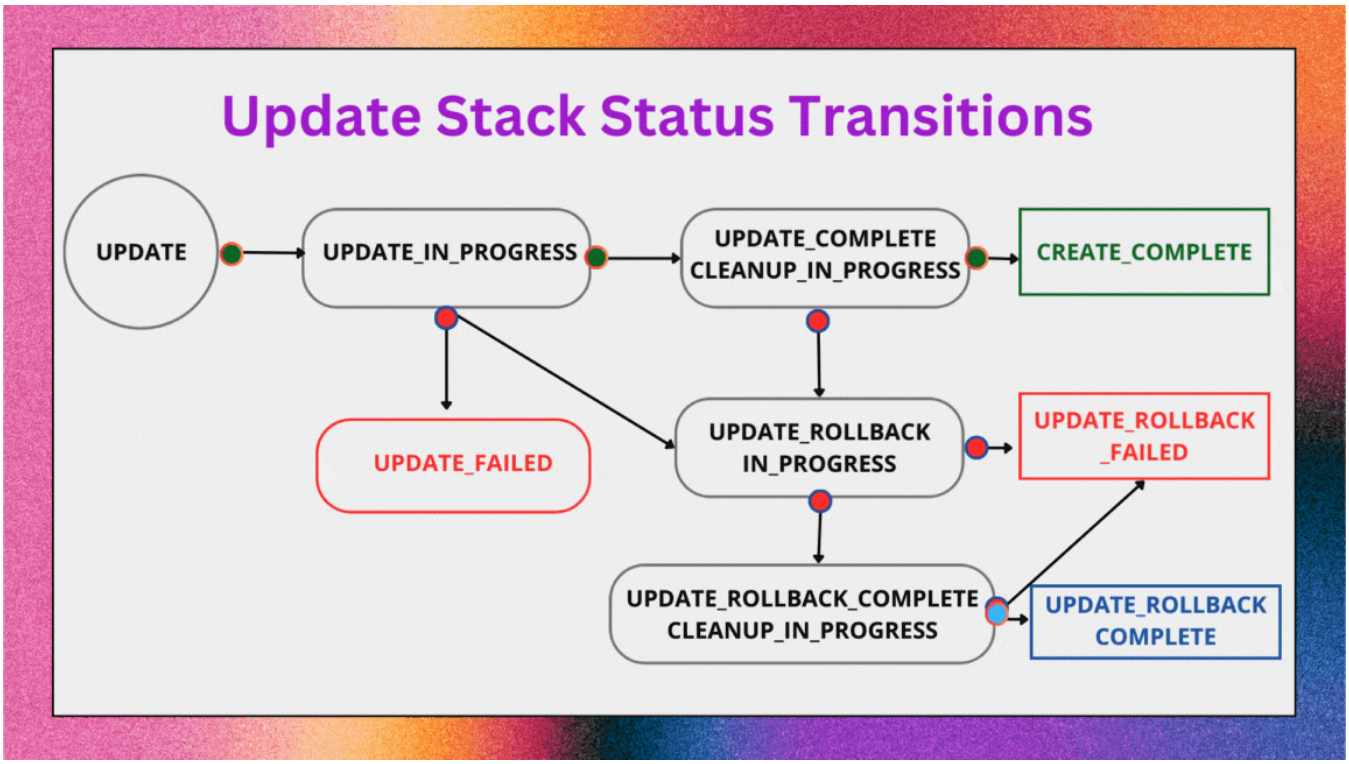
📌 Below flow diagram below will explain in detail about create stack status transitions.



👉 When you trigger stack creation, you will first see the CREATE_IN_PROGRESS status.
👉 If the creation process completes without errors, the stack will reach the CREATE_COMPLETE status.
👉 If there are invalid parameter values, insufficient permissions, or some other initial error, the stack will reach the CREATE_FAILED status. Inspect the stack Events and correct any issues before retrying.
👉 If there are errors after some resources have been created, the stack will attempt to roll back(ROLLBACK_IN_PROGRESS) and remove the new resources.
- If the rollback is successful (ROLLBACK_COMPLETE), you will need to delete the stack before retrying.
👉 If the rollback fails (ROLLBACK_FAILED), review the stack Events and fix any issues before retrying the rollback process or deleting the stack.

🌈 **Update Stack Status Transitions:**

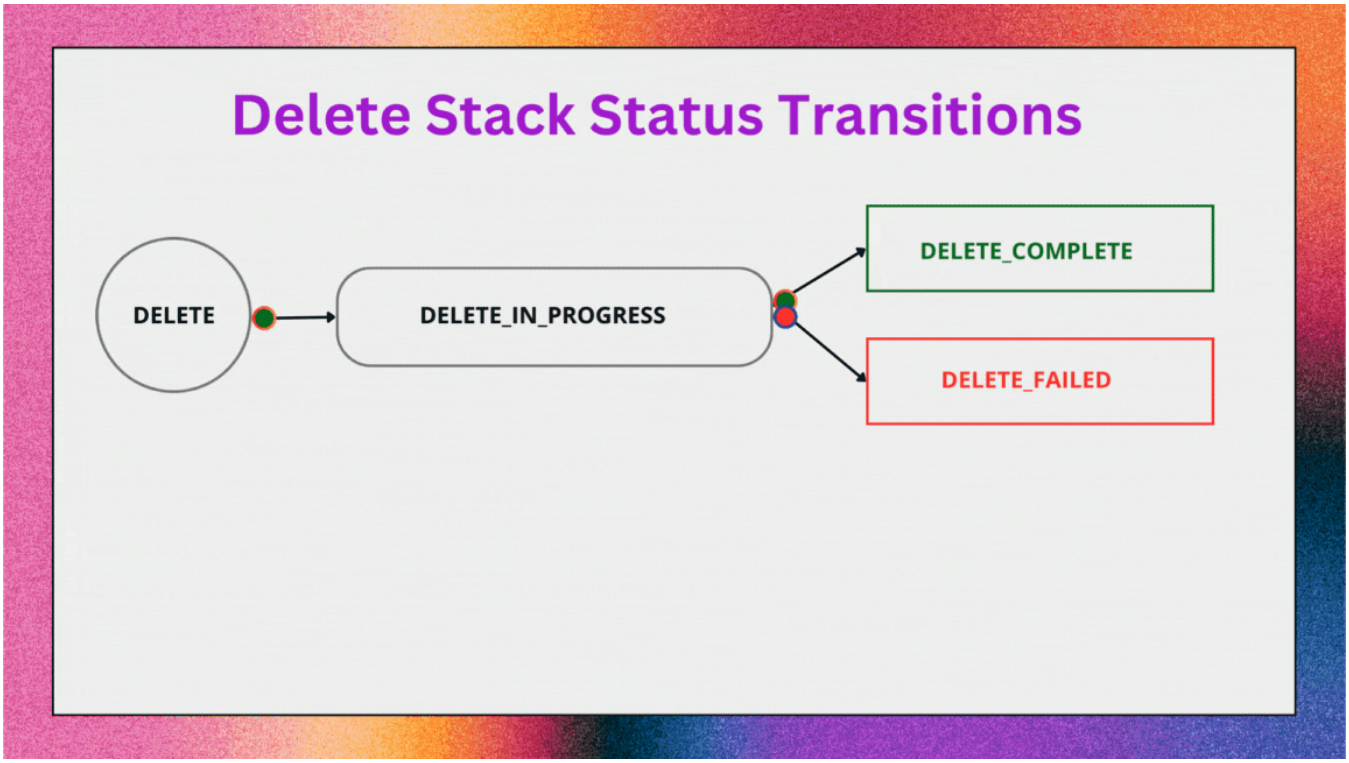
📌 Below flow diagram below will explain in detail about update stack status transitions.



👉 When you trigger a stack update, you will first see the UPDATE_IN_PROGRESS status.
👉 If there are errors in the update process, the stack will reach the UPDATE_FAILED status. You will need to review the stack Events before retrying.
👉 If the desired changes are applied without errors, CloudFormation will perform a cleanup (remove any old resources) and the stack will get to the UPDATE_COMPLETE_CLEANUP_IN_PROGRESS status.
👉 If cleanup is successful, the stack will get to the UPDATE_COMPLETE status.
👉 If cleanup fails, it will attempt to roll back the changes (UPDATE_ROLLBACK_IN_PROGRESS).
👉 If rollback fails, the stack will get to the UPDATE_ROLLBACK_FAILED status. Review the stack Events and fix any issues before retrying the rollback process or deleting the stack.
👉 If rollback succeeds, it will also trigger a cleanup, and the stack will get to the UPDATE_ROLLBACK_COMPLETE_CLEANUP_IN_PROGRESS status.
👉 If the after-rollback cleanup is successful, you will get to the UPDATE_ROLLBACK_COMPLETE status. That is, your stack will get to its initial state. You should review the Events before retrying.
👉 If the after-rollback cleanup fails, the stack will get to the UPDATE_ROLLBACK_FAILED status. Review the stack Events and fix any issues before retrying the rollback process or deleting the stack.

🌈 Delete Stack Status Transitions:

📌 Below flow diagram below will explain in detail about delete stack status transitions.



👉 When you trigger a stack deletion, you will first see the DELETE_IN_PROGRESS status.
👉 If the deletion process completes without errors, the stack will reach the DELETE_COMPLETE status.
👉 If deletion goes wrong, the stack will reach the DELETE_FAILED status. Inspect the stack Events and correct any issues before retrying.

🌈 Summary:

📌 In the above blog, We discussed in detail about the creation of EC2 Instances using the CloudFormation stack and also discussed the various statuses that a CloudFormation stack can have and the transitions that occur as the stack moves from one status to another.

🕵️ I also want to express that your feedback is always welcome. As I strive to provide accurate information and insights, I acknowledge that there's always room for improvement. If you notice any mistakes or have suggestions for enhancement, I sincerely invite you to share them with me.

> *🤩 Thanks for being patient and following me. Keep supporting 🙏*
>
> *Clap 👏 if you liked the blog.*
>
> *For more exercises — please follow me below ✅!*
>
> *https://medium.com/@vjraghavanv*

#aws #cloudformation #stack #template #transition #ec2 #awsreskill #awscommunitybuilder #machinelearning #ec2instances #awsugmdu #awsugncr #automatewithraghavan

AWS | Cloudformation | Ec2 | Cloud Computing | Stack

Follow

Written by Vijayaraghavan Vashudevan

103 followers · 74 following

Hi Everyone !! Am here to publish the technical topics for the community which includes cloud concepts, postman, automation, RPA etc. Please do follow me :)

No responses yet